Linear Regression (LR)

Definition

→ Linear regression assumes linear relationship b/w input and output variables.

→ It takes data and finds out an equation of straight line that best fits the data.

Notations in LR

m = no of training examples

x's = input variables | features

y's = output | target variable

Hypothesis

$$h_\theta(x) = \theta_0 + \theta_1(x)$$

θ_0 : Intercept , θ_1 : slope

Linear Regression with One Variable (Univariate)

→ In this case, we have one input parameter and on the basis of it, we predict output.

$$h(x) = \theta_0 + \theta_1 (x)$$

Now how would we determine θ_0 and θ_1 ?

we have following three ways of finding θ_0 and θ_1.

Method of Least Squares

→ In this method, we minimize Sum of Squares of Errors (SSE)

$$SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

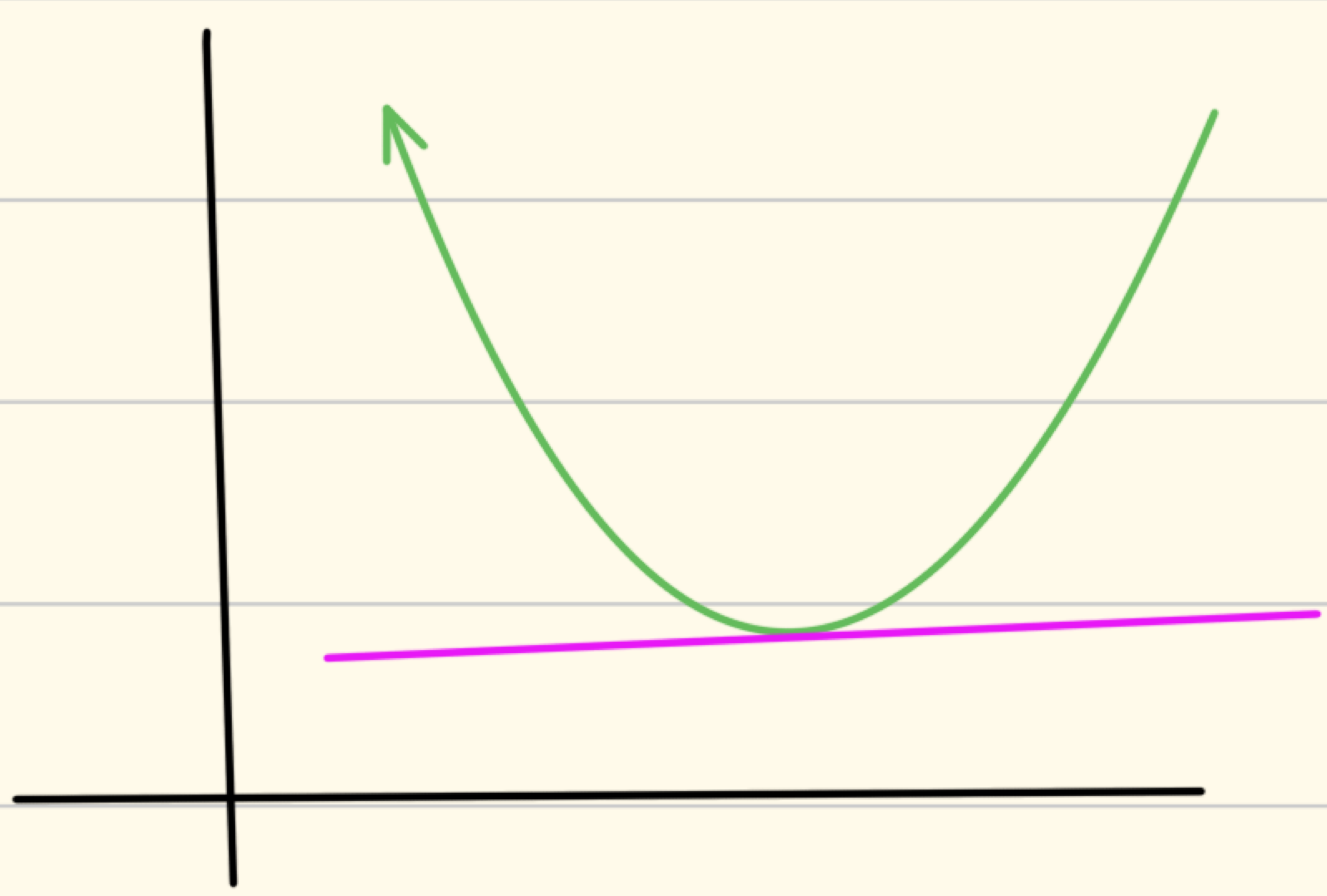
$$= \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

$\because\ \theta_0 \rightarrow b_0 \quad , \quad \theta_1 \rightarrow b_1$

→ we want to minimize SSE, so we would find where derivative is zero.

→ derivative $= 0$, gives local minima of a function.

e.g



→ So, let's find derivatives with respect to b_0 and b_1.

$$\frac{d(SSE)}{d(b_0)} :$$

$$\frac{d(SSE)}{d(b_0)} = \frac{d}{d b_0}\left(\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2\right)$$

Applying power rule here

$$= 2\left(\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)\right)(-1)$$

$$\frac{d\,SSE}{d\,b_0} = -2\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)$$

putting $\frac{d\,SSE}{d\,b_0} = 0$

$$0 = -2\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)$$

$$\frac{0}{-2} = \sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)$$

$$\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i) = 0$$

opening brackets and applying summation

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} b_0 - b_1 \sum_{i=1}^{n} x_i = 0$$

$$\sum_{i=1}^{n} y_i - n b_0 - b_1 \sum_{i=1}^{n} x_i = 0$$

$$\Rightarrow$$

$$n b_0 = \sum_{i=1}^{n} y_i - b_1 \sum_{i=1}^{n} x_i$$

$$\boxed{b_0 = \frac{\sum_{i=1}^{n} y_i - b_1 \sum_{i=1}^{n} x_i}{n}}$$

we can also write it as:

$$b_0 = \frac{\sum_{i=1}^{n} y_i}{n} - b_1 \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\boxed{b_0 = \bar{y} - b_1 \bar{x}}$$

$$\frac{d(SSE)}{d(b_1)}:$$

$$\frac{d(SSE)}{db_1} = \frac{d}{db_1}\left(\sum_{i=1}^{n}(y_i - b_0 - b_1x_i)^2\right)$$

$$= 2\sum_{i=1}^{n}(y_i - b_0 - b_1x_i)(-x_i)$$

putting $\frac{dSSE}{db_1} = 0$

$$0 = \sum_{i=1}^{n} x_iy_i - b_0\sum_{i=1}^{n}x_i - b_1\sum_{i=1}^{n}x_i^2$$

$$b_1\sum_{i=1}^{n}x_i^2 = \sum_{i=1}^{n}x_iy_i - b_0\sum_{i=1}^{n}x_i$$

putting value of b_0 and representing $\sum_{i=1}^{n}$ as simple Σ

$$b_1 \sum x_i^2 = \sum x_i y_i - \frac{(\sum y_i - b_1 \sum x_i)}{n}(\sum x_i)$$

$$b_1 \sum x_i^2 = \frac{n \sum x_i y_i - (\sum y_i - b_1 \sum x_i)(\sum x_i)}{n}$$

$$n b_1 \sum x_i^2 = n \sum x_i y_i - (\sum y_i)(\sum x_i) + b_1 (\sum x_i)^2$$

$$n b_1 \sum x_i^2 - b_1 (\sum x_i)^2 = n \sum x_i y_i - (\sum y_i)(\sum x_i)$$

$$b_1 \left(n \sum x_i^2 - (\sum x_i)^2\right) = n \sum x_i y_i - (\sum x_i)(\sum y_i)$$

$$\boxed{b_1 = \frac{n \sum_{i=1}^{n} x_i y_i - (\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n \sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}}$$

$$b_1 = \frac{\sum_{i=1}^{n} \left((x_i - \bar{x})(y_i - \bar{y})\right)}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

First find $(x_i - \bar{x})$ column,

then find $(y_i - \bar{y})$ column,

then find $(x_i - \bar{x})(y_i - \bar{y})$ column

then find $\Sigma\left((x_i - \bar{x})(y_i - \bar{y})\right)$

Method of Normal Eq

→ We want to minimize SSE

$$SSE = \sum_{i=1}^{n} (y_i - (b_0 + b_1 x_i))^2$$

→ write it into matrix form.

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_3 \end{bmatrix}, \quad \beta = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

$$Y = X\beta$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_3 \end{bmatrix} \times \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

$$= \begin{bmatrix} b_0 + b_1 x_1 \\ b_0 + b_1 x_2 \\ \vdots \\ b_0 + b_1 x_n \end{bmatrix}$$

represents predicted values

Writing SSE in matrix form:

$$SSE = \sum (y_i - \hat{y}_i)^2$$

$$= (Y - X\beta)^T (Y - X\beta)$$

$\because$ multiplying a vector with its transpose gives Sum of Squares.

e.g:

$$e = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}, \quad e^T = \begin{bmatrix} e_1 & e_2 \end{bmatrix}$$

$$e^T e = e_1^2 + e_2^2$$

That's why we wrote:

$$SSE = (Y - X\beta)^T (Y - X\beta)$$

$$= Y^T Y - Y X^T \beta^T - Y^T X \beta + X^T \beta^T X \beta$$

$$\because \; Y^T X \beta = \beta^T X^T Y$$

$$= Y^T Y - \beta^T X^T Y - \beta^T X^T Y + \beta^T X^T X \beta$$

$$= Y^T Y - 2\beta^T X^T Y + \beta^T X^T X \beta$$

As we want to minimize, so let's take derivate w.r.t β

∵ Rules

$$\frac{d}{d\beta}(\beta^T A \beta) = 2A\beta$$

$$\frac{d}{d\beta}(c^T\beta) = c$$

- $$\frac{d}{d\beta}\left(\underbrace{-2X^TY}, \beta^T\right) = -2X^TY$$
- $$\frac{d}{d\beta}\left(\beta^T \underbrace{X^TX}_{A} \beta\right) = 2X^TX\beta$$

$\Rightarrow$

$$\frac{dSSE}{d\beta} = 0 - 2X^TY + 2X^TX\beta$$

putting $\frac{dSSE}{d\beta} = 0$

$$\Rightarrow \quad 2X^TX\beta = 2X^TY$$

$$X^TX\beta = X^TY$$

$$X^T Y - X^T X \beta = 0$$

$$X^T (\underbrace{Y - X\beta}_{\text{error vector}}) = 0$$

$$X^T e = 0$$

$\Rightarrow$ dot product of X^T (features) and error vector (e) is zero, so the e vector is $\perp$ to features.

That's why it's called normal eq. of simple LR.

Solving for β

$$X^T X = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$$

$$= \begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$= \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}$$

$\Rightarrow$

$$\begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}$$

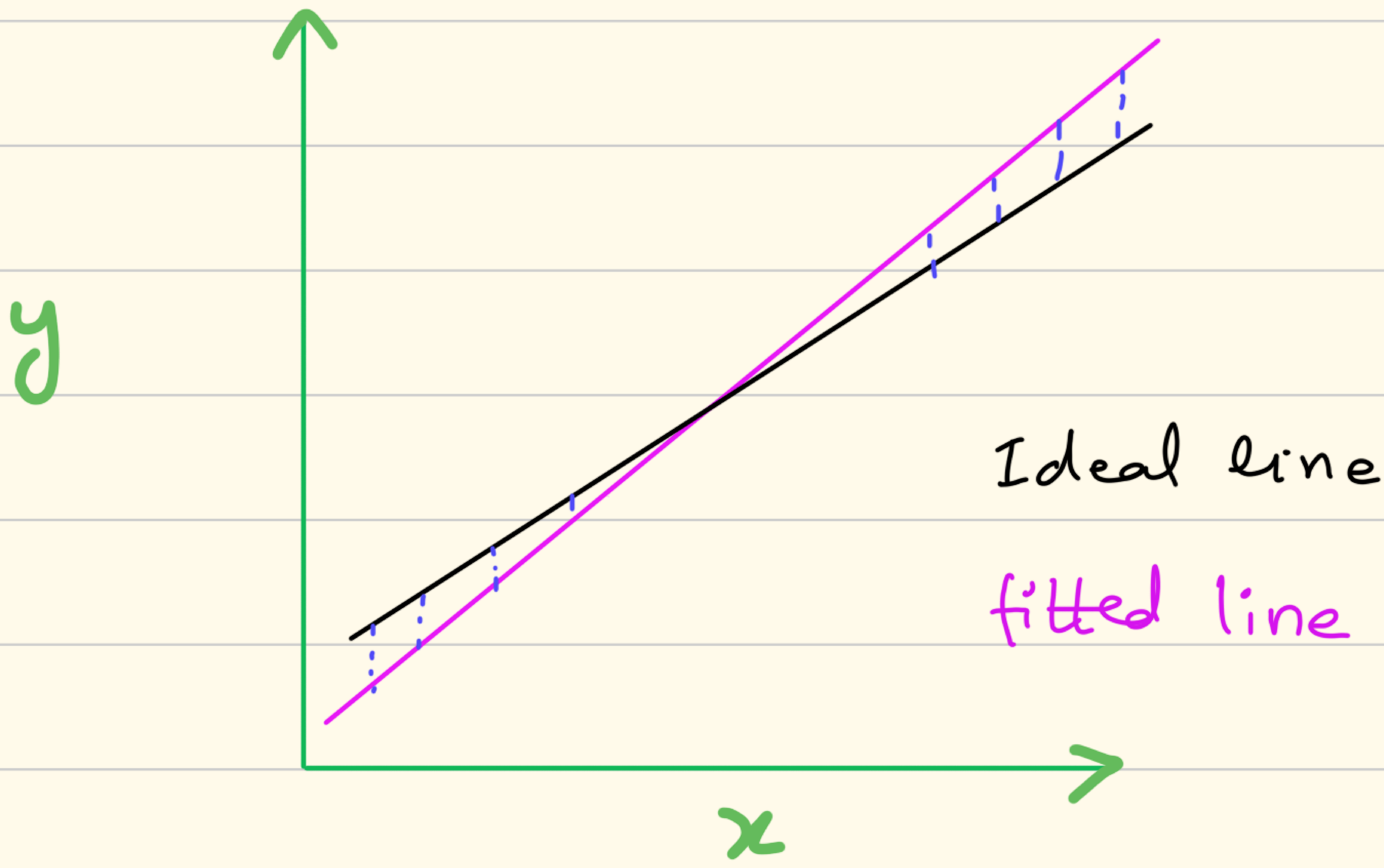
$$n b_0 + b_1 \Sigma x_i = \Sigma y_i$$

$$b_0 \Sigma x_i + b_1 \Sigma x_i^2 = \Sigma x_i y_i$$

→ we will solve in same way as we did in "Least Squares Method" and we will get same equations for b_0 and b_1.

(Simpler Version)

We say that, errors are $\perp$ to input values.



In matrix notation, we write input feature vector as

$$X = \begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{n1} \\ \vdots & & & & \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix}$$

x_{ij} show ith instance of jth feature.

⇒ each column shows one feature.

now feature vectors are placed in columns, so we can say that

$$X^T . e = 0$$

{ X.e won't represent that error vector is perendicular to features }

we can write it as:

$$X^T (Y - X\beta) = 0$$

$$X^T Y - X^T X \beta = 0$$

$$X^T X \beta = X^T Y$$

$$\beta = (X^T X)^{-1} (X^T Y)$$

Now we can solve it to get β matrix.

Solving for β

$$X^T X = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$$

$$= \begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$= \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}$$

$\Rightarrow$

$$\begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}$$

$$n b_0 + b_1 \Sigma x_i = \Sigma y_i$$

$$b_0 \Sigma x_i + b_1 \Sigma x_i^2 = \Sigma x_i y_i$$

→ we will solve in same way as we did in "Least Squares Method" and we will get same equations for b_0 and b_1.

Method of Gradient Descent

→ Here we minimize cost function $J(\theta_0, \theta_1)$

→ here instead of b_0 and b_1 we are using θ_0, θ_1

$$J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^i) - y^i\right)^2$$

(mean square errors)

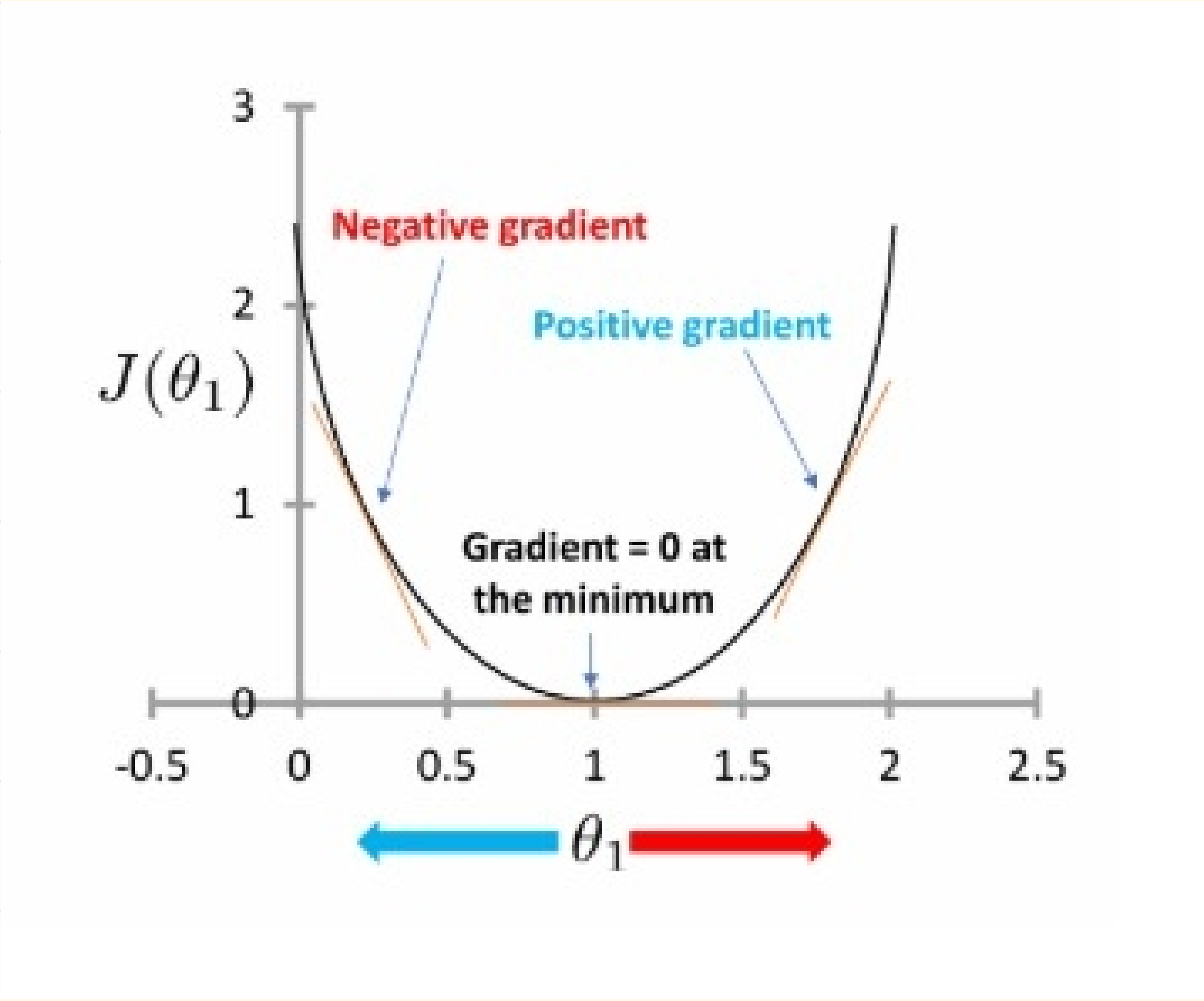
In future calculation, we would get '2' in numerator, so to make terms clean we append '2' in denominator which creates no impact.

$\Rightarrow$

$$J(\theta_0,\theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^i) - y^i)^2$$

→ we have to minimize $J(\theta_0, \theta_1)$

so we take derivatives w.r.t θ_0 and θ_1



Algorithm

1. Start with some θ_0 and θ_1

②

$$\text{temp0} = \theta_0 - (LR) \times \text{slope}$$

$$= \theta_0 - (\alpha) \times \frac{d}{d\theta_0} J(\theta_0, \theta_1)$$

$$= \theta_0 - (\alpha) \times \frac{d}{d\theta_0} \left(\frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^i) - y^i \right)^2 \right)$$

$$= \theta_0 - (\alpha) \times \frac{d}{d\theta_0} \left(\frac{1}{2m} \sum_{i=1}^{m} \left(\theta_0 + \theta_1 x^i - y^i \right)^2 \right)$$

$$\text{temp1} = \theta_1 - (LR) \times \text{Slope}$$

$$= \theta_1 - (\alpha) \times \frac{d}{d\theta_1} J(\theta_0, \theta_1)$$

$$= \theta_1 - (\alpha) \times \frac{d}{d\theta_1} \left(\frac{1}{2m} \sum_{i=1}^{m} \left(\theta_0 + \theta_1 x^i - y^j \right) \right)$$

$$\theta_0 = \text{temp0}$$

$$\theta_1 = \text{temp1}$$

→ here we are updating values of θ_0 and θ_1

→ Note that we are using previous values of θ_0 and θ_1 while current calculations.

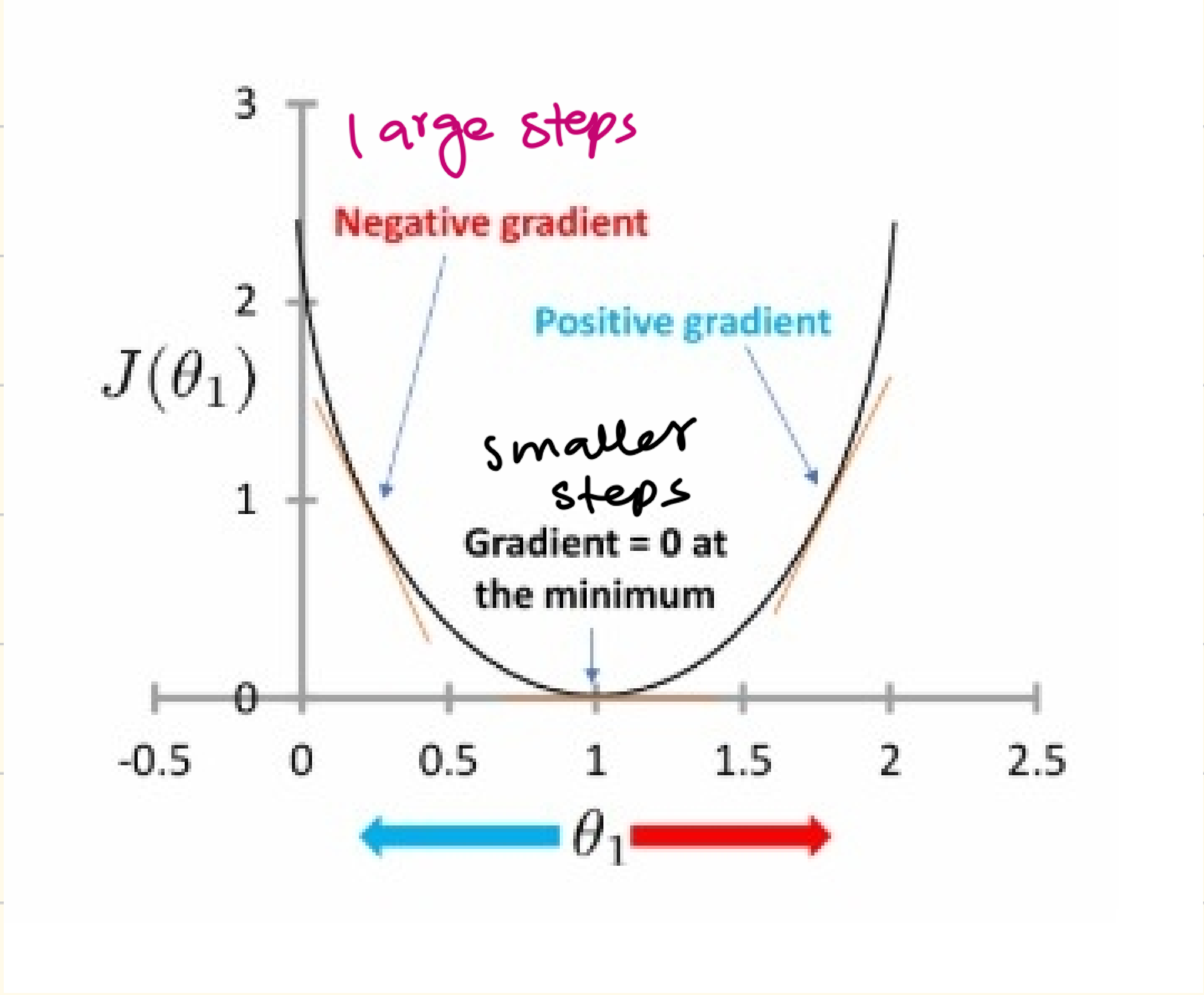
3 Repeat step #02 until convergence (means when values of θ_0 and θ_1 become too small)

Incorrect:

$$
\begin{aligned}
\text{temp0} &:= \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) \\
\theta_0 &:= \text{temp0} \\
\text{temp1} &:= \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) \\
\theta_1 &:= \text{temp1}
\end{aligned}
$$

→ Gradient descent takes large steps initially as slope is large at start.

Then takes small steps when slope of loss function would be less.



Calculation of $Slope_{\theta_0}$ & $Slope_{\theta_1}$

$Slope_{\theta_0}$

$$\frac{d}{d\theta_0} J(\theta_0, \theta_1) = \frac{d}{d\theta_0} \left(\frac{1}{2m} \sum_{i=1}^{n} (\theta_0 + \theta_1 x^i - y^i)^2 \right)$$

$$= \frac{1}{\cancel{2}m} \times \cancel{2} \sum_{i=1}^{m} (\theta_0 + \theta_1 x^i - y^i)(1)$$

$$\boxed{\frac{d}{d\theta_0} J(\theta_0, \theta_1) = \sum_{i=1}^{m} (\underbrace{\theta_0 + \theta_1 x^i}_{h(x_i)} - y^i)}$$

$Slope_{\theta_1}$:

$$\frac{d}{d\theta_1} J(\theta_0, \theta_1) = \frac{d}{d\theta_1} \left(\frac{1}{2m} \sum_{i=1}^{m} (\theta_0 + \theta_1 x^i - y^i)^2 \right)$$

$$= \frac{1}{\cancel{2}m} \times \cancel{2} \sum_{i=1}^{m} (\theta_0 + \theta_1 x^i - y^i)(x^i)$$

$$\frac{d}{d\theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^{m} (\theta_0 + \theta_1 x^i - y^i) \times x^i$$

Generally:

$$\boxed{\frac{d}{d\theta_j} J(\theta_0, \dots, \theta_n) = \frac{1}{m} \sum_{i=1}^{m} \left(\theta_0 + \theta_j x^i - y^i\right) \times x_j^i}$$

Types of Gradient Descent

1- Batch GD

→ m == n

→ weights (θ_j(s)) are updated after whole data processing.

→ means first slope for all data point (x^i, y^i) is found and summed, then value to weights is updated.

$$\theta_j = \theta_j - (LR) \times \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^i) - y^i \right) \times x_j^i$$

2- Stochastic GD

→ Here weights are updated after finding slope for every point (x^i, y^i)

→ In next step, updated values of weights would be used.

$$\theta_j = \theta_j - (LR) \times \left(h_\theta(x) - y \right) \times (x)$$

3- Mini-Batch GD

→ Values of weights are updated after processing of a batch size (size is predetermined).

$$\theta_j = \theta_j - \alpha \times \frac{1}{b} \sum_{i=1}^{b} \left(h_\theta(x_j^i) - y^i \right) \times x_j^i$$

$\therefore \quad 1 < b < m$

Example

(x) x₁ (Size)	(y) # y₁ (Price)
800	150
900	170
1000	190
1100	210
1200	230
1300	250
1400	270
1500	290
1600	310
1700	330

Initial Values:

$$\theta_0 = 10$$
$$\theta_1 = 0.5$$
$$LR = 0.00001$$

Calculation of $(\theta_0 + \theta_1 x_i)$, $(\theta_0 + \theta_1 x_i - y_i)$ and $(\theta_0 + \theta_1 x_i - y_i) \times x_i$

→ we would compute these terms to put in the formula of Slope_{θ_0} & Slope_{θ_1}

$\overbrace{(\theta_0 + \theta_1 x^i)}^{h(x_i)}$	$(\theta_0 + \theta_1 x^i - y^i)$	$(\theta_0 + \theta_1 x^i - y^i) x^i$
410	260	208000
460	290	261000
510	320	320000
560	350	385000
610	380	456000
660	410	533000
710	440	616000
760	470	705000
810	500	800000
860	530	901000
	(3950)	(5185000)

$$\frac{d}{d\theta_0} = \frac{1}{m} \sum_{i=1}^{m} (\theta_0 + \theta_1 x^i - y^i)$$

$$= \frac{1}{10} \times 3950 = 395$$

$$\frac{d}{d\theta_1} = \frac{1}{m} \sum_{i=1}^{m} (\theta_0 + \theta_1 x^i - y^i) \times x^i$$

$$= \frac{1}{10} \times 5185000 = 518500$$

Now let's find new θ_0 and θ_1

$$\theta_0 = (10) - (0.00001) \times (395)$$
$$\theta_0 = 9.99605$$

$$\theta_1 = (0.5) - (0.00001) \times (518500)$$
$$\theta_1 = -4.685$$

Linear Regression with Multiple Variables (Multi-variate)

$$J(\theta_0, \theta_1, \ldots, \theta_n) = \frac{1}{2m} \sum_{i=1}^{m} (h(x^i) - y^i)^2$$

repeat {

$$\theta_j = \theta_j - (\alpha) \times \frac{d}{d\theta_j} J(\theta_0, \ldots, \theta_n)$$

$$\forall \; j = 0, 1, \ldots, n$$

} until convergence